NSO8062 // NSO7007 Geofüüsikaline andmetöötlus

Loeng 1: Sissejuhatus

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Loengute aeg ja materjalid

- Kas neljapäeviti 10:00-13:15 on OK?
- Materjalid ilmuvad MOODLE-sse. Kas teil on ligipääs?
- Andmed panen Google Drive-i. Kas teil on konto?
- Kes kindlalt soovib seda ainet võtta?

Töö käik

- Igas loengus teeme harjutusi, lahendame ülesandeid ja töötleme ning analüüsime etteantud keskkonnaalaseid andmeid.

- Iga lahendatud ülesande kohta valmib tudengil skript, mis tuleb õppejõule esitada vähemalt päev enne järgmist loengut.

- Iga loengu kohta saab tudeng punkte vastavalt esitatud skriptidele. Punkte saab kui skript on esitatud õigeks ajaks, skript töötab ja annab õige vastuse.

Hindamine

- Iga harjutus annab max. 3 punkti.
- Lõpphinne koosneb
  - semestri jooksul kogutud punktidest (50%) ja
  - eksamitulemusest (50%).
- Kui semestri jooksul on punkte kogutud alla 20% või on eksamitulemus alla 20%, siis ei ole lõpphinne positiivne.

Hinne	Hinde ülemine piir (%, >)	Hinde alumine piir (%, <=)
5	100	84
4	84	68
3	68	52
2	52	36
1	36	20
0	20	0

Harjutused - üldiselt

- **Harjutusi teeme MATLAB keskkonnas**.
- Analüüsime aegridu (1D), ruumiandmeid (2D) ja aja ning ruumiandmeid (ND)
- Analüüsid sisaldavad: erinevate statistikute leidmist (miinimum, maksimum, keskmised, standardhälve, standardviga, protsentiilid), regressiooni analüüsi (lineaarne, polünomiaalne, korrelatsioon, p-väärtus), mudelandmete võrdlemist seireandmetega jne.
- Andmeid loeme sisse ja salvestame erinevates formaatides (.mat, .txt, .csv, .xlsx, .nc)

Harjutused - üldiselt

- Õpime tegema jooniseid. Erinevad joonised profiilidest ja aegridadest, tuulteroosid, tuulevektorite joonised, histogrammid, kaardid, videod, tulpdiagrammid jms.
- Õpime nö tühimikke täitma – andmeid interpoleerima (nt. lineaarselt, polünomiaalselt).
- Analüüsitavad keskkonna parameetrid: tuulekiirus, tuulesuund, õhutemperatuur, veetemperatuur, soolsus, hapnik, veetase, vee läbipaistvus, üldlämmastik, üldfosfor, nitrit+nitraat, ammonium, fosfaat, sügavus jne.

Tudengid

- Mis õppekavad?
- Kas olete andmetöötlusega varem kokku puutunud?
  - JAH: Milliste ainete raames? Mida teinud olete? Mis keeltes?
- Mida ootate antud ainelt? Nt. seoses teie teiste ainetega, lõputööga, huviga jne.

Ajakava ja teemad

Nädal	Teema	Õppejõud
1	Sissejuhatus ainesse	S-T. Luik
2	Erinevad andmete tüübid ja lihtsamad tegevused nendega.	S-T. Luik
3	NetCDF failide loomine, muutmine jne.	S-T. Luik
4	Keri andmed. Profiilide visualiseerimine, andmete eemaldamine, silumine, gradientide leidmine.	S-T. Luik
5	Andmete 1D interpoleerimine ja korrelatsioonid, statistikud jne.	S-T. Luik
6	Tuuleandmed ja nende visualiseerimine	S-T. Luik
7	2D andmed, interpoleerimine, visualiseerimine (nt kaart)	
8	TBD	
9	TBD	
10	TBD	
11	TBD	
12	TBD	
13	TBD	
14	TBD	
15	Konsultatsioon	
16	**Eksam**	
17		
18		
19	10/06/2026 kevadsemestri eksamite sooritamise lõpptähtaeg	
20	15/06/2026 kevadsemestri lõpp	

MATLAB tarkvara

- TalTech IT kasutajatoe lehel otsi “MATLAB” või:
- https://taltech.atlassian.net/servicedesk/customer/portal/14/article/38995352?source=search
- Vastavalt juhistele tuleb teha MathWorks lehel konto, laadida alla tarkvara ja logida sisse tehtud kontoga.
- Installeeri toolboxid ->

```
-----------------------------------------------------------------------------------------------------
MATLAB Version: 24.2.0.2923080 (R2024b) Update 6
MATLAB License Number: 40687166
Operating System: Microsoft Windows 11 Education Version 10.0 (Build 26200)
Java Version: Java 1.8.0_202-b08 with Oracle Corporation Java HotSpot(TM) 64-Bit Server VM
mixed mode
-----------------------------------------------------------------------------------------------------
MATLAB                                                Version 24.2       (R2024b)
Curve Fitting Toolbox                                 Version 24.2       (R2024b)
Database Toolbox                                      Version 24.2       (R2024b)
Financial Instruments Toolbox                         Version 24.2       (R2024b)
Financial Toolbox                                     Version 24.2       (R2024b)
Mapping Toolbox                                       Version 24.2       (R2024b)
Navigation Toolbox                                    Version 24.2       (R2024b)
Optimization Toolbox                                  Version 24.2       (R2024b)
Signal Processing Toolbox                             Version 24.2       (R2024b)
Statistics and Machine Learning Toolbox               Version 24.2       (R2024b)
Text Analytics Toolbox                                Version 24.2       (R2024b)
```

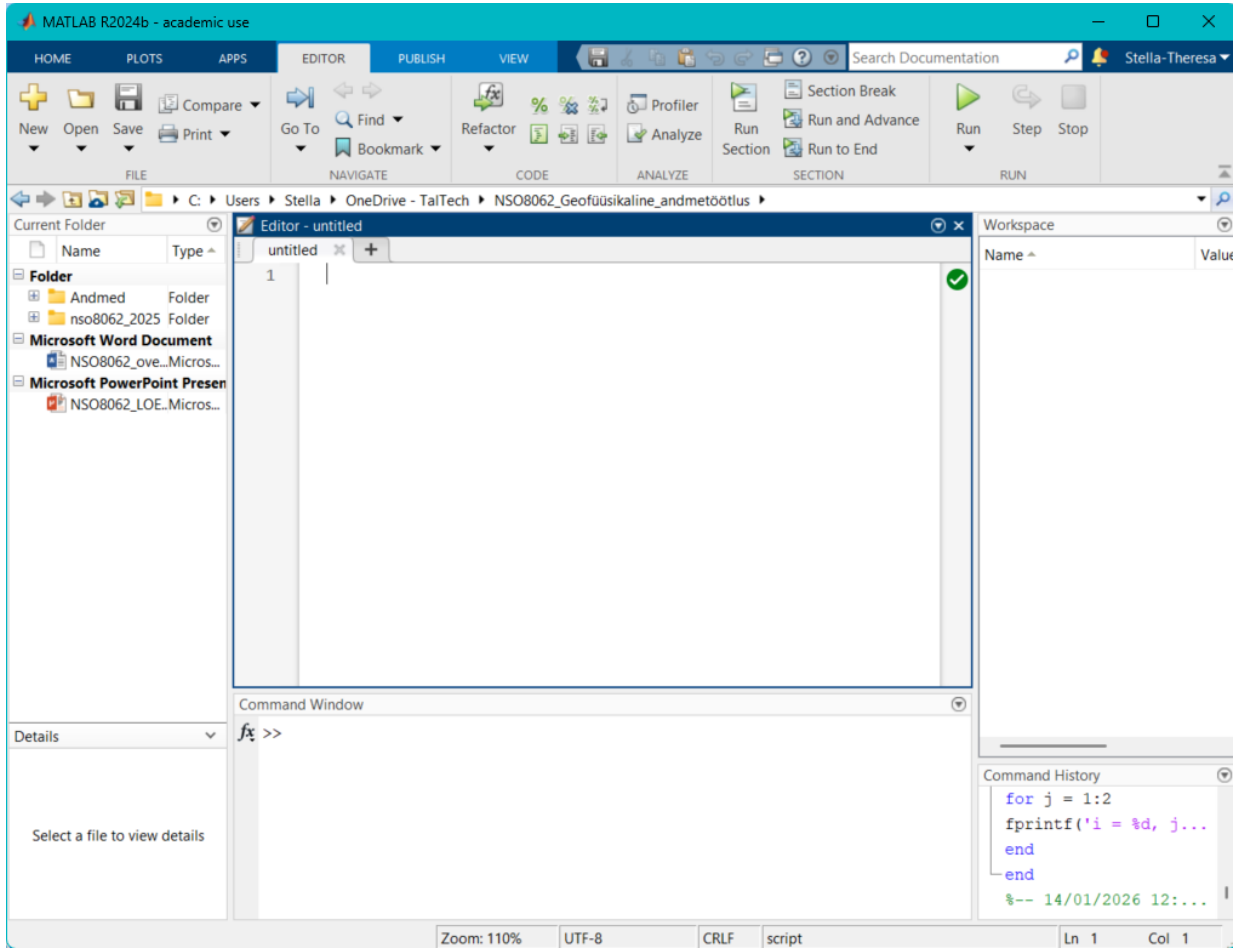
Analüüsitavad andmed - https://drive.google.com/drive/folders/1INWLh0tS1toZrJ_Sp3YHIE5t1w-Df0Tg?usp=sharing

- ERA5 hourly data on single levels from 1940 to present (https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview . 10m u-component of wind, 10m v-component of wind, Sea surface temperature, Land-sea mask. North: 66°, West: 9°, South: 53°, East: 30°. 2016-2025, 12:00. .nc)
- Keri jaama profiilid (2020, .mat)
- Mudelandmed (põhjakihi ja setete andmed, veesamba andmed, 2020, .nc)
- ICES seireandmed (eeltöödeldud ajaloolised andmed 1940 - , .csv)
- Estmodel jõgede sissevoolu andmed (.mat)
- EMODnet batümeetria andmed (.nc)

Sissejuhatus MATLABi (Matrix Laboratory)

Recommended layout:

- Command Window
- Current Folder
- Details
- Editor
- Workspace
- Command History



Sissejuhatus MATLABi

Teksti peale % käsitletakse kommentaarina.

clc – puhastab Command Window

Ctrl+Enter jooksutab skripti või kahe %% vahel oleva skripti osa.

Skripti alustame:

- clearvars – teeb Workspace'i puhtaks
- close all – paneb jooniseid kinni nt
- format long – komakohtade arv, mida näidatakse
- clc

Sissejuhatus MATLABi

- Entering commands into the command window

  2+3

- Creating variables
  - Variable Naming Conventions: Variable names must start with a letter, followed by letters, numbers, or underscores. For example, myVariable_1 is valid, while 1stVariable is not.

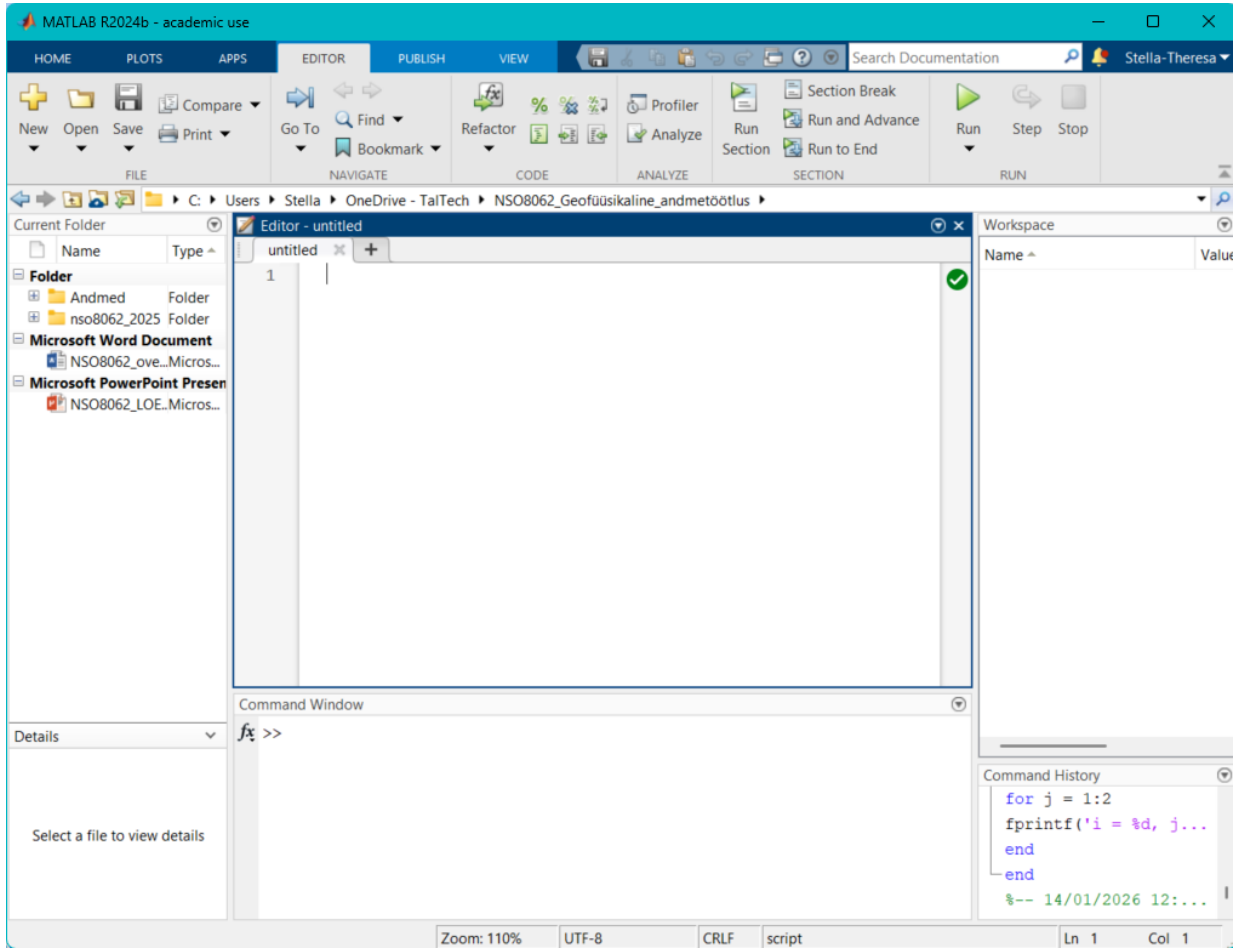
  a = 2;

  b = 3;

  c = a + b

- Displaying Variable Values: Use the disp function or just type the variable name.

  disp(c)



MATLAB Data types

- Numeric Data Types
  - Scalars, Vectors (1D), Matrices (2D), multidimensional arrays (ND):

    A = [1, 2, 3; 4, 5, 6]; % 2x3 matrix
- Logical Data Type

    isTrue = (5 > 3); % Returns true (1)
- Character Arrays and Strings
  - Creating and Manipulating Character Arrays:

    str = 'Hello, MATLAB'!;

    strArray = ["Hello", "World"];

MATLAB Data types

- Container Data Types
  - cell - Cell arrays for storing different data types. Elements accessed using curly braces {}

    ```
    myCell = {1, 'text', [1 2 3]; pi, eye(2), 'MATLAB'}
    myCell{1,2}  % returns 'text’
    ```

  - struct - Structure arrays for organizing related data. Fields accessed using dot notation

    ```
    student.name = ‘Jaan';
    student.id = 12345;
    student.grades = [4 3 5];
    ```

  - table - Database-like container. Labeled rows and columns.

    ```
    T = table([1;2], ['A';'B'], {'x';'y'}, 'VariableNames', {'ID', 'Grade', 'Name'})
    ```

MATLAB Data types

- Categorical, datetime, duration, etc.

- Tips for Working with Data Types
  - Check type with class(): **class(x)**
  - Convert between types with explicit functions: **double(), char(), string()**, etc.
  - Check array size with size(): **size(A)**
  - Use **whos A** to display variables in workspace and their types

MATLAB Arrays and Matrices

- Creating Arrays

```
Z = zeros(3, 3); % 3x3 matrix of zeros
O = ones(2, 4);  % 2x4 matrix of ones
R = rand(4, 4);  % 4x4 matrix of random numbers
L = linspace(0, 10, 5); % Generates [0 2.5 5 7.5 10]
```

- Indexing and Slicing
  - Accessing Elements:

```
A(1, 2) % Accesses the element in the 1st row, 2nd column
```

  - Modifying Elements:

```
A(1, 2) = 10; % Changes the value at (1, 2) to 10
```

MATLAB Arrays and Matrices

- Matrix Operations
  - Basic Operations:

```
A = [2, 3; 4, 5];
B = [1, 2; 4, 1];
C = A + B; % Matrix addition
D = A * B; % Matrix multiplication
E = A .* B; % Element-wise multiplication
F = A';    % Transpose of A
```

MATLAB Conditional statements

```
x = 10;

if x > 5
        disp('x is greater than 5');
elseif x == 5
        disp('x is equal to 5');
else
        disp('x is less than 5');
end
```

MATLAB Loops – for, while

```
for i = 1:5
        disp(i);
end
```

```
count = 1;
while count <= 5
        disp(count);
        count = count + 1;
end
```

MATLAB Loops – nested loops

```
for i = 1:3
        for j = 1:2
                fprintf('i = %d, j = %d\n', i, j);
        end
end
```

- %d is a format specifier for displaying integers (decimal notation)
- \n is an escape sequence that creates a new line after printing

Feature	Floating-Point (double)	Integer (int32, uint8, etc.)
Default	Yes, the MATLAB standard.	Must be explicitly cast.
Decimal Support	Yes.	No (decimals are rounded/truncated).
Memory Usage	High (8 bytes for double).	Low (1 to 8 bytes).
Best Use Case	Scientific math, engineering.	Loop counters, image pixels, hardware.
Mathematical Precision	Finite (subject to "eps").	Exact for whole numbers.

Operator	Type of Data	Example Output
%d	Integer (Decimal)	42
%f	Floating-point (Fixed-point)	42.000000
%s	String or Character array	"Hello"
%e	Exponential notation	4.2000e+01

MATLAB Functions – create your own

```
function result = addNumbers(a, b)
        result = a + b;
end
```

Saves it as a function (.m).
Call upon it: addNumbers(var1,var2)

Example:

```
var1 = 7;
var2 = 4;
result = addNumbers(var1,var2)
result =
   11
```

MATLAB Functions – built-in

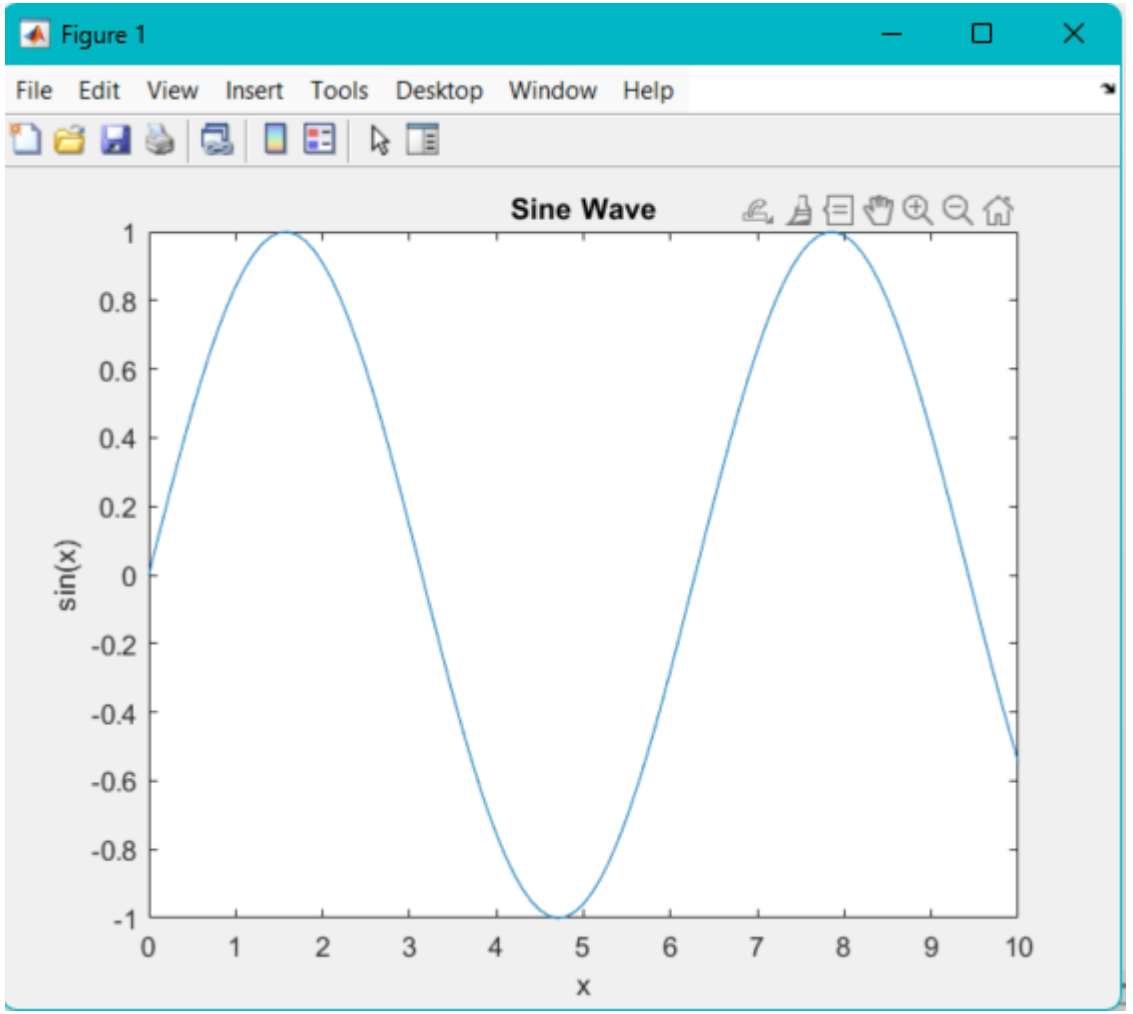
```
sumValue = sum([1, 2, 3, 4]); % Sum of elements
avgValue = mean([1, 2, 3, 4]); % Average
maxValue = max([1, 2, 3, 4]); % Maximum value
…

square = @(x) x.^2; % Anonymous function for squaring
result = square(5);  % Returns 25
…
```

MATLAB – Data visualization

- Plotting basics
  - Creating 2D plots (plot, scatter, line …)

```
x = 0:0.1:10;  % from 0, by 0.1, to 10
y = sin(x);
plot(x, y); % Basic sine wave plot
title('Sine Wave');
xlabel('x');
ylabel('sin(x)');
```

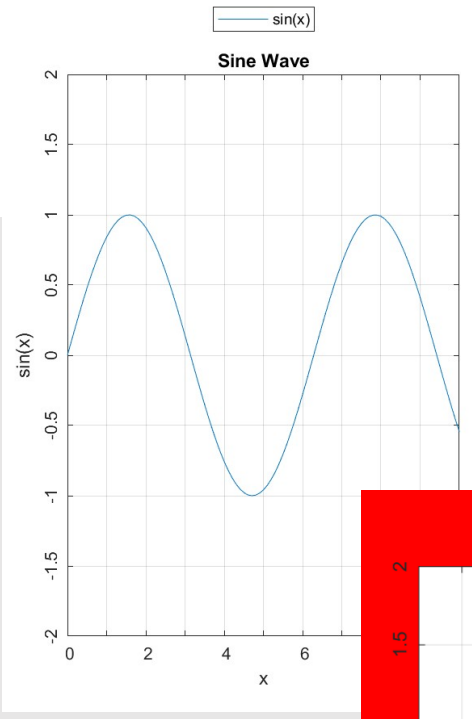


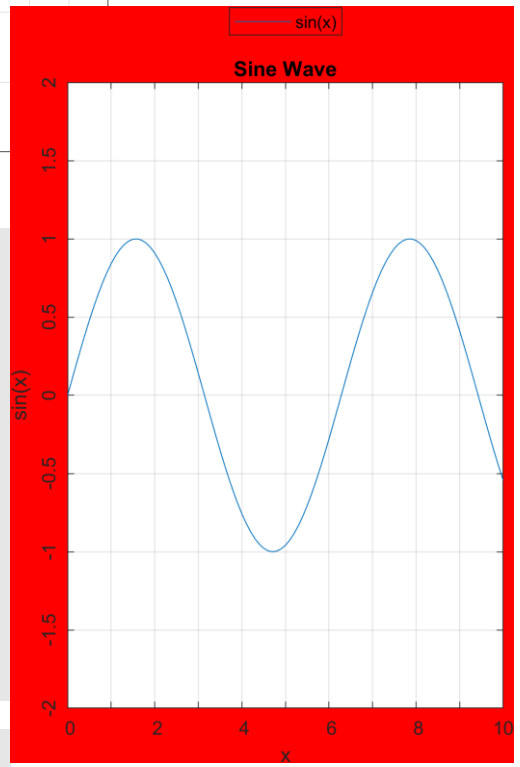
MATLAB – Data visualization

- Modify figure

```
% Set all x-ticks from 0 to 10 with interval of 1
xticks(0:1:10);
% Create labels for every other x-tick (0:2:10)
all_ticks = 0:1:10;
labels = cell(size(all_ticks));
labels(1:2:end) = cellstr(num2str(all_ticks(1:2:end)')); % Labels for positions 0,2,4,6,8,10
labels(2:2:end) = {''}; % Empty labels for positions 1,3,5,7,9
% Apply the labels
xticklabels(labels);
```

```
% y-tick angle
ytickangle(90)
% Specify axis limits
ylim([-2 2])
% Modify figure size
g = gcf;
g.Position = ([10 10 400 600]);
% Add grid
grid on;
% Add legend
legend('sin(x)','Location','northoutside','Color','none');
```






```
saveas(gca,'example_figure_v1.jpg')
exportgraphics(gcf, 'example_figure_v2.png', 'Resolution', 300, 'BackgroundColor', 'red');
```

MATLAB – File Input and Output

```
data = load('datafile.txt'); % Load data from a text file
tbl = readtable('datafile.csv'); % Read CSV file into a table
data = ncread(filename, variableName); % Get .nc data
```

```
save('outputfile.mat', 'data'); % Save variables to a .mat file
writetable(tbl, 'outputfile.csv'); % Write table to CSV
writematrix(A,'A.txt','Delimiter','tab’); % Write matrix to a text file
```